Effects of a 12-Week Aerobics Exercise Program on Mental Health Among Obese College Students: A Randomized Controlled Trial

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Scope Statement

This manuscript fits well within the scope of Frontiers in Public Health, particularly the specialty section related to Public Mental Health, Physical Activity, and Health Promotion. The study investigates the effects of a 12-week aerobics exercise intervention on mental health outcomes among obese college students using a randomized controlled trial design. Obesity and mental health problems among young adults have become important global public health concerns, especially within university populations experiencing increasing psychological stress and sedentary lifestyles. This research provides empirical evidence that structured aerobic exercise may improve psychological well-being and mental health indicators in obese college students. The findings contribute to the growing body of public health literature examining non-pharmacological and lifestyle-based interventions for improving mental health in youth populations. In addition, the study has practical implications for university health promotion programs, obesity prevention strategies, and exercise-based mental health interventions in higher education settings.

Conflict of interest statement

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest

Credit Author Statement

YANG WANG: Data curation, Formal Analysis, Funding acquisition, Investigation, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xiaoxia Yan**: Formal Analysis, Investigation, Software, Visualization, Writing – original draft. **Zainal Abidin Bin Zainuddin**: Conceptualization, Resources, Supervision, Validation, Writing – review & editing.

Keywords

Aerobics exercise, BMI, college students, Mental Health, Obesity, randomized controlled trial, WHR

Abstract

Word count: 140

Obesity among college students is a rising public health concern, often accompanied by mental health issues such as anxiety, depression, and low self-esteem. This study evaluated the impact of a 12-week structured aerobics exercise program on mental health indicators in obese college students. A randomized controlled trial involving 150 participants from Yuncheng University, China, compared three groups: Control (no intervention), Experimental Group 1 (self-selected activities), and Experimental Group 2 (structured aerobics exercise).Results showed that Experimental Group 2 had significant reductions in anxiety and depression scores and improvements in overall mental well-being compared to other groups. Regression analysis indicated BMI and WHR as predictors of mental health outcomes like paranoia and dependence. Findings underscore the effectiveness of structured aerobics exercise over self-selected activities, suggesting its integration into university health strategies. Future studies should explore long-term impacts and compare various exercise modalities.

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Studies involving animal subjects

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Studies involving human subjects

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No Generative AI was used in the preparation of this manuscript.

Effects of a 12-Week Aerobics Exercise Program on Mental Health Among Obese College Students: A Randomized Controlled Trial

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ABSTRACT:Obesity among college students is a rising public health concern, often accompanied by mental health issues such as anxiety, depression, and low self-esteem. This study evaluated the impact of a 12-week structured aerobics exercise program on mental health indicators in obese college students. A randomized controlled trial involving 150 participants from Yuncheng University, China, compared three groups: Control (no intervention), Experimental Group 1 (self-selected activities), and Experimental Group 2 (structured aerobics exercise).Results showed that Experimental Group 2 had significant reductions in anxiety and depression scores and improvements in overall mental well-being compared to other groups. Regression analysis indicated BMI and WHR as predictors of mental health outcomes like paranoia and dependence. Findings underscore the effectiveness of structured aerobics exercise over self-selected activities, suggesting its integration into university health strategies. Future studies should explore long-term impacts and compare various exercise modalities.

KEY WORDS: Obesity, aerobics Exercise, Mental Health, College Students, Randomized Controlled Trial, BMI, WHR

Introduction

Obesity has emerged as a global public health crisis, with its prevalence reaching alarming rates across diverse populations(1, 2). According to the World Health Organization (WHO), more than 1.9 billion adults worldwide are overweight, and over 650 million of these individuals are classified as obese(3, 4). This epidemic is no longer confined to developed nations; developing countries, including China, are experiencing a significant rise in obesity rates, especially among younger demographics such as college students(5). Obesity not only contributes to a range of physical health complications, including cardiovascular diseases and diabetes, but also profoundly affects mental health(6, 7).

Among college students, obesity has been linked to mental distress, including heightened levels of anxiety, depression, and low self-esteem(8). The college years represent a critical period of life when individuals face academic pressures, social challenges, and identity formation(9). For obese students, these stressors are often exacerbated by body image issues, social stigmatization, and reduced physical activity, leading to a vicious cycle of poor mental health and limited coping mechanisms(10, 11). Thus, addressing the mental impacts of obesity in this population is a pressing concern for researchers, healthcare providers, and educators alike(12).

Interventions targeting mental health in obese college students are of paramount importance, as untreated mental distress can hinder academic performance, disrupt social relationships, and adversely affect overall quality of life(13). Among the array of potential interventions, physical exercise has gained significant attention as a non-pharmacological and cost-effective strategy for promoting both physical and mental well-being(14). Specifically, aerobic exercise, characterized by sustained rhythmic activities that improve cardiovascular fitness, has been shown to reduce symptoms of anxiety and depression while enhancing self-esteem and emotional resilience(15). Its accessibility and adaptability make aerobics exercise an ideal intervention for college students, who often have limited resources and time constraints(16).

While numerous studies have explored the effects of exercise on mental health, few have focused specifically on structured aerobic exercise programs for obese college students(16). Existing research often targets general populations or individuals with clinical mental health conditions, overlooking the unique challenges and needs of obese young adults in an academic setting(17). Moreover, the majority of studies lack a randomized controlled trial (RCT) design, limiting the ability to draw robust conclusions about causality and effectiveness.

In the Chinese context, despite the growing prevalence of obesity among college students, there remains a dearth of empirical research addressing the dual impacts of obesity on physical and mental health(18). Structured aerobic exercise programs, which are both evidence-based and culturally adaptable, have yet to be thoroughly evaluated for their mental health benefits in this specific population. These gaps underscore the need for well-designed, population-specific studies to inform intervention strategies and policy development.

The present study aims to address these gaps by evaluating the effects of a 12-week aerobics exercise program on mental health indicators among obese college students. Using a randomized controlled trial design, this study examines changes in key mental outcomes, including anxiety, depression, and self-esteem, following participation in the intervention(19). By focusing on a culturally relevant and accessible exercise modality, this research seeks to provide actionable insights for integrating aerobics exercise into mental health promotion programs in higher education settings. Additionally, the study contributes to the growing body of literature on the intersection of physical and mental health, with implications for obesity management and mental well-being in young adults(20).This study not only seeks to validate the mental health benefits of aerobics exercise but also highlights its potential as a scalable intervention for tackling the dual burdens of obesity and mental distress in college students.

Methods

Participants

This study involved 150 obese college students recruited from [Yuncheng University, Shanxi, China]. Participants were selected based on the following inclusion criteria:Age between 18 and 22 years.Body mass index (BMI) ≥ 28 kg/m² and waist-to-hip ratio (WHR) ≥ 0.90 for males or ≥ 0.85 for females, meeting the criteria for obesity as defined by the World Health Organization (WHO) and widely accepted anthropometric standards(21).Enrollment as full-time undergraduate students during the study period.No history of psychiatric disorders or current treatment for mental conditions. No contraindications to moderate-intensity physical exercise as determined by a medical screening.

The exclusion criteria included:Pregnancy or lactation.Use of medications affecting body weight or mental conditions within the past six months.Participation in structured exercise programs or weight management interventions in the three months preceding the study.

Participants were stratified by gender to ensure equal representation of males and females across groups. They were then randomly assigned into three groups:Control group (n = 50): No specific intervention was provided. Participants were instructed to maintain their usual routines.Experimental Group 1 (n = 50): Participants engaged in self-selected physical activities such as walking or cycling, with a target of achieving 5,000 steps per session, three times per week.Experimental Group 2 (n = 50): Participants participated in a structured aerobics exercise program, detailed below.

Intervention

The structured aerobics exercise program in Experimental Group 2 was designed to provide a standardized and reproducible intervention.

Duration: 12 weeks.Frequency: Three sessions per week.Session Length: Each session lasted 60 minutes, including 10 minutes of warm-up, 40 minutes of moderate-intensity aerobics exercise, and 10 minutes of cool-down.

Exercise Modalities: Activities included step aerobics, rhythmic movements, and low-impact dance routines tailored to participants' fitness levels.

Intensity: Moderate-intensity, corresponding to 50–70% of the participants' maximum heart rate, monitored using heart rate monitors.

Participants in Experimental Group 1 were instructed to engage in self-selected physical activities of similar frequency and duration but without the structured intensity or supervision provided in Experimental Group 2. Both experimental groups were provided with a diary to record their activity adherence, and weekly checks were conducted by research staff to ensure compliance.

Assessment Tools

Mental measures were used to assess the participants' mental health status at three time points: baseline (Week0), mid-intervention (Week 6), and post-intervention (Week12). The following validated scales were administered:The Chinese College Student Mental Health Scale (CCSMHS)(22, 23).This scale, developed by Professor Zheng Richang of Beijing Normal University, assesses 12 dimensions of mental health, including Somatization；Anxiety；Depression；Inferiority；Paranoid；Compulsion；Social withdrawal；Social aggression；Sexual psychology；Dependence；Impulse；Psychosis.Higher scores indicate poorer mental health(24).All assessments were conducted in a quiet, controlled environment to ensure the accuracy and reliability of responses. Researchers were trained in the administration of the questionnaires and were available to clarify any participant questions.

Data Collection Points

Data were collected at three intervals:Baseline (Week 0), Before the intervention. Mid-Intervention(Week6),To evaluate interim progress.Post-Intervention (Week 12),To d etermine the overall effects of the intervention.Adherence to the intervention was track ed using attendance logs (Experimental Group 2) and step-counting devices (Expe rimental Group 1).

Statistical Analysis

Data were analyzed using SPSS 27.0 and the following statistical methods were employed:Descriptive Statistics;One-Way ANOVA;Repeated Measures ANOVA;Correlati on Analysis;Multiple Regression Analysis. Significance Level: A p< 0.05 was considered statistically significant. Effect sizeswere calculated to assess the magnitude of observed effects(25).

Results

Baseline Characteristics

Demographic Information of Respondents

By detailing the number of participants in the study, we can gain an initial understanding of the basic characteristics of the sample, providing a foundation for subsequent analysis. Among the 150 participants, there were 75 males and 75 females, with a gender ratio of 1:1. The balanced distribution of gender helps to reduce bias caused by gender differences, making the study results more generalizable and representative. Table1 shows the general statistics of the research objects:

TABLE1 General statistics of the subjects.

Content	Control Group	Experimental Group 1	Experimental Group 2
Sample size	50	50	50
1-A1 Age	19.34±0.56	19.50±0.58	19.50±0.54
1-A2 Whether there are any siblings①0 ②1 ③2 ④3 or more	①44 ②6	①45 ②5	①45 ②5
1-A3Are parents obese28?(Obesity:weight/height2 ≧28kg/cm2)① None ② either parent is overweight ③ Both parents are overweight	①10 ②40	①9 ②41	①7 ②43
A4 Major① Liberal arts ② science ③ engineering ④ Medicine ⑤ art and sports ⑥ other __	① 16②13 ③ 15④ 5	① 9② 13 ③ 18④ 3 ⑥ 7	① 5② 23 ③ 13⑥ 9
1-A5 Family location ① urban ② rural	① 43 ②7	① 46 ② 4	① 44② 6
A6 The average monthly household income is ① 3000 ② 3000-5000 ③6000- 10,000 ④ ≧ 10,000	① 1 ② 37③ 10 ④ 2	② 39 ③ 9 ④ 2	② 42③ 7 ④ 1
2-B1 Choose your current body outline from the figure above?	① 1 ② 1 ③ 2 ④ 31 ⑤ 7 ⑥ 7 ⑦1	②2 ③ 2 ④ 32 ⑤ 6 ⑥ 8	② 3 ③ 2 ④ 19⑤ 14 ⑥ 12 ⑦1
2-B2 Choose your ideal body shape from the figure above?	① 2② 4 ③ 14④ 16 ⑤ 8 ⑥ 6	①1 ② 2 ③ 13④ 18 ⑤ 12⑥ 4	① 1② 2 ③ 12④ 23 ⑤9 ⑥ 3
3-C1 The intensity of your usual physical exercise is: ① Light exercise ② small intensity but not too tense sports ③ Moderate intensity intense sustained exercise ④ shortness of breath, sweating a lot in intense, but not lasting sports ⑤ Shortness of breath, sweating a lot in the intensity of sustained exercise	① 1 ② 22 ③ 16 ④ 1 ⑤ 1	9 ② 13 ③ 18 ④ 9 ⑤1	18 ② 11 ③ 14 ④ 7
3-C2 How many minutes at a time do you do the above intensity physical activities ：	① 20 ② 6	① 16 ② 8	① 1② 24 ③ 18④ 5

① Under 10 minutes ②11-20 minutes ③ about 30 minutes ④40-60 minutes ⑤ more than 60 minutes	③ 16 ④ 8	③ 22 ④ 4	⑤ 2
3-C3 How many times a month do you take part in the above sports activities： ① Once at most ②2-3 times ③ 1-2 times a week ④ 3-5 times a week ⑤ about once a day	① 1② 3 ③ 34④ 9 ⑤ 3	② 4③ 40 ④ 6	② 10③ 33 ④ 5 ⑤ 2

Starting from the demographic characteristics and exercise behaviors, The following outlines the preliminary analysis steps and conclusions based on these data.

In terms of age, the control group had an average age of 19.34±0.56, Experimental Group 1 had 19.50±0.58, and Experimental Group 2 had 19.50±0.54. The average ages across all groups were similar, with small standard deviations, indicating no significant age differences among the groups. Most participants in each group had either 0 or 1 sibling, with very few having 2 or more siblings.The proportion of parents with obesity was slightly higher in Experimental Group 2 compared to the control group and Experimental Group 1, but the overall distribution was fairly close.There were notable differences in the distribution of majors, particularly in the fields of engineering, medicine, and arts and physical education. The differences in major distribution among the groups could potentially influence their health and exercise behavior habits. Most participants were from urban areas, with fewer from rural areas. The urban-rural distribution was fairly consistent across all groups, with the majority being urban residents. The household monthly income for the control group and the experimental groups was concentrated between 3,000-5,000 and 6,000-10,000, showing similar income distribution across the groups.

The current body shape is mostly chosen between ④ (31-32 participants) and ⑤ (6-7 participants), but more participants in Experimental Group 2 selected ⑤ and ⑥. This suggests that participants in Experimental Group 2 might have higher expectations for their body shape. The ideal body shape is mostly selected as ③, ④, or ⑤, indicating that their ideal body shape is close to medium or slightly slim. The ideal body shape among the participants is relatively consistent, with little difference between the groups. The exercise intensity in the control group, Experimental Group 1, and Experimental Group 2 is generally consistent, focusing on moderate to low-intensity exercise. The duration of each exercise session for the control group, Experimental Group 1, and Experimental Group 2 is concentrated in the range of ② 11-20 minutes and ③ about 30 minutes. The frequency of exercise per month among participants in the control group, Experimental Group 1, and Experimental Group 2 is similar, typically around 1-2 times per week.

Analysis of Differences in Pre-Experiment Data

Before commencing the 12-week experimental study, a difference analysis of the baseline data was conducted to ensure that the initial conditions of each group were balanced.

This was done to determine whether there were any significant differences between the control and experimental groups at the start of the experiment. Independent samples t-tests and Analysis of Variance (ANOVA) were used to statistically examine the obesity indicators, physical health indicators, and mental health indicators across the groups. Table2 presents the results of the differential analysis of the obesity evaluation indicators, physical, and mental test indicators before the experiment:

TABLE2 Analysis of Differences in BMI, WHR, and Test Indicators.

Content	Indicators	Control group (Mean ± SD)	Experimental group 1 (Mean ± SD)	Experimental group2 (Mean ± SD)	F-Value	P-Value
Obesity evaluation	BMI	32.13±1.52	32.20±1.52	32.30±1.60	0.138	0.871
	WHR	0.94±0.03	0.94±0.03	0.94±0.03	0.171	0.843
Mental health test	Somatization	16.02±1.84	16.08±1.88	16.24±1.87	0.186	0.830
	Anxiety	17.64±2.24	17.84±2.00	17.88±1.88	0.197	0.821
	Depression	18.00±2.47	18.28±1.89	17.68±2.61	0.819	0.443
	Inferiority	18.16±2.20	18.62±3.33	18.02±2.42	0.678	0.509
	Paranoid	19.66±2.06	20.54±2.71	20.34±2.98	1.563	0.213
	Compulsion	18.76±1.71	19.18±2.22	18.88±1.95	0.604	0.548
	Social withdrawal	16.22±2.70	15.64±2.41	15.96±2.03	0.735	0.481
	Social aggression	16.40±4.05	16.12±3.08	16.06±2.08	0.164	0.849
	Sexual psychology	12.96±2.95	13.12±2.05	12.74±2.38	0.320	0.727
	Dependence	18.84±1.87	18.28±2.50	18.82±1.66	1.212	0.301
	Impulse	16.02±2.75	16.02±2.90	16.38±2.66	0.281	0.755
	Psychosis	11.52±1.42	11.46±1.23	11.28±1.31	0.446	0.641

The results of the difference analysis showed that there were no significant differences (p-values all greater than 0.05) in the obesity indicators, and mental test indicators among the groups before the experiment. This indicates that the random grouping effectively balanced the initial conditions across the groups, providing reliable baseline data for assessing the intervention effects in the subsequent experiment. By ensuring consistency in the initial conditions, the impact of the 12-week aerobicss exercise on the physical and mental health of obese college students can be evaluated more accurately.

Group Comparisons Across Time Points

Mid-Test Analysis of Mental Health Indicators

At the mid-point of the 12-week intervention, mental health indicators were assessed across the three groups: Control Group, Experimental Group 1 (self-selected physical activities), and Experimental Group 2 (moderate-intensity aerobicss). The following analysis explores the differences between these groups, focusing on the mental outcomes and their comparison to the national average level.The data were analyzed using One-Way ANOVA to determine the statistical significance of differences between these groups. The results of this analysis are presented below in Table 3:

TABLE 3 One-Way ANOVA Results for Mental Health Indicators at Mid-Test.

Indicators	Control Group Mean ±SD	Experimental Group 1 Mean ±SD	Experimental Group 2 Mean ±SD	F-Value	P-Value	National Average Level
Somatization	15.84±1.75	15.08±1.41	14.62±1.32	8.349	<0.001	13.27±4.24
Anxiety	17.14±1.62	16.94±2.34	16.18±2.31	2.865	0.060	16.60±5.36
Depression	17.76±1.88	17.16±2.20	16.68±2.18	3.348	0.038	16.00±5.37
Inferiority	17.70±1.53	17.08±1.87	16.22±3.18	5.200	0.007	15.12±5.15
Paranoia	19.20±3.00	17.82±1.99	17.32±2.00	8.373	<0.001	15.13±4.95
Force	18.56±2.12	18.22±1.97	17.88±2.18	1.319	0.271	18.23±5.32
Social Withdrawal	16.06±2.08	15.36±2.51	15.00±2.02	2.964	0.055	15.59±5.37
Social Attack	16.32±2.03	15.26±2.81	14.88±3.11	3.841	0.024	14.27±4.38
Psychosexual	12.64±2.07	12.46±2.53	11.46±1.80	4.366	0.014	11.37±4.01
Dependence	18.66±1.72	18.02±2.20	17.00±2.09	8.641	<0.001	16.78±5.45
Impulse	15.86±2.10	15.22±2.11	15.30±2.71	1.124	0.328	14.72±4.49
Psychosis	11.50±1.40	10.88±1.44	10.60±1.63	4.757	0.010	12.00±3.65

The mid-test mental indicators reveal that the structured aerobicss program in Experimental Group 2 has had a positive impact on several key mental health domains, including somatization, depression, inferiority, paranoia, and psychosis. While some indicators, such as anxiety and social withdrawal, did not reach statistical significance, the trends suggest that continued participation in structured physical activities could further reduce these symptoms. Overall, the findings indicate that regular, moderate-intensity exercise can be an effective intervention for improving mental health, though some areas may require longer-term or more targeted interventions to align with national averages.

Post-Test Analysis of Mental Health Indicators

At the conclusion of the 12-week intervention, the mental health indicators were measured across the three groups: Control Group, Experimental Group 1 (self-selected physical activities), and Experimental Group 2 (moderate-intensity aerobicss). The following analysis delves into the differences observed among these groups, with a focus on the effectiveness of the interventions in comparison to the national average levels.The results of this analysis are presented below in Table 4:

TABLE 4 One-Way ANOVA Results for Mental Health Indicators at Post-Test.

Indicators	Control	Experimental	Experimental	F-Value	P-Value	National

	Group Mean ±SD	Group 1 Mean ±SD	Group 2 Mean ±SD			Average Level
Somatization	15.42±1.65	13.28±1.75	12.79±1.86	31.482	<0.001	13.27±4.24
Anxiety	17.66±1.38	16.02±2.17	15.02±1.82	26.802	<0.001	16.60±5.36
Depression	17.10±1.66	15.92±2.39	14.82±2.20	14.666	<0.001	16.00±5.37
Inferiority	17.58±2.11	14.90±1.75	14.52±2.57	29.496	<0.001	15.12±5.15
Paranoia	18.68±2.93	16.38±1.72	15.08±1.72	34.291	<0.001	15.13±4.95
Force	18.68±1.95	17.46±1.77	16.14±2.54	18.044	<0.001	18.23±5.32
Social Withdrawal	15.80±2.15	15.20±2.49	14.04±1.80	8.553	<0.001	15.59±5.37
Social Attack	15.98±1.90	14.42±2.56	13.32±2.05	18.598	<0.001	14.27±4.38
Psychosexual	12.64±2.03	11.32±2.39	10.24±1.57	17.654	<0.001	11.37±4.01
Dependence	17.90±1.61	16.46±2.04	15.62±2.67	16.771	<0.001	16.78±5.45
Impulse	15.22±2.18	14.42±2.05	14.12±2.40	3.291	<0.001	14.72±4.49
Psychosis	11.40±1.43	10.32±1.45	10.20±1.39	10.814	<0.001	12.00±3.65

Summary of Mid-Test and Post-Test Analyses

Across both the mid-test and post-test stages, Experimental Group 2 (moderate-intensity aerobicss) consistently outperformed the Control Group and Experimental Group 1 in nearly all mental health indicators. The structured aerobics exercise program demonstrated significant benefits in reducing symptoms associated with somatization, anxiety, depression, inferiority, paranoia, obsessive-compulsive tendencies, social withdrawal, aggression, psychosexual issues, dependence, impulsivity, and psychosis. By the end of the 12-week intervention, participants in Experimental Group 2 not only showed improvement compared to the Control Group but also, in many cases, achieved mental health outcomes that were better than the national average.

The results highlight the effectiveness of structured, moderate-intensity physical exercise in promoting mental health and reducing various mental symptoms. The improvements observed in Experimental Group 2 suggest that regular engagement in structured physical activity can lead to substantial mental health benefits, underscoring the importance of incorporating such programs into mental health interventions. The consistency of these findings from the mid-test to the post-test stages further supports the sustainability of these benefits over time.

Correlations Between Exercise Adherence and mental Outcomes

The results shown in Table5 highlight the correlation between a 12-week aerobics exercise program and mental health in different groups.

TABLE 5 Correlation Between aerobicss Exercise and Mental Health Indicators.

Indicator	Group	Pearson Correlation Coefficient (r) ——Significance (Two-tailed) (p)			Sample Size (N)
		Pre-Mid	Pre-Post	Mid-Post	

		r	p-value	r	p-value	r	p-value	
Somatization	Control Group	0.147	0.309	0.051	0.725	0.769	<0.001	50
	Experimental Group 1	-0.187	0.194	-0.081	0.575	0.602	<0.001	50
	Experimental Group 2	-0.251	0.078	-0.232	0.105	0.457	<0.001	50
Anxiety	Control Group	-0.251	0.079	0.012	0.932	0.47	<0.001	50
	Experimental Group 1	-0.128	0.375	0.057	0.694	0.783	<0.001	50
	Experimental Group 2	-0.197	0.170	-0.100	0.488	0.542	<0.001	50
Depression	Control Group	-0.097	0.505	-0.010	0.945	0.722	<0.001	50
	Experimental Group 1	0.255	0.074	0.123	0.395	0.737	<0.001	50
	Experimental Group 2	-0.040	0.784	-0.138	0.340	0.451	0.001	50
Inferiority	Control Group	-0.174	0.228	-0.214	0.136	0.523	<0.001	50
	Experimental Group 1	0.008	0.955	-0.202	0.159	0.662	<0.001	50
	Experimental Group 2	-0.120	0.406	-0.107	0.461	0.610	<0.001	50
Paranoid	Control Group	-0.038	0.792	-0.100	0.491	0.944	<0.001	50
	Experimental Group 1	-0.065	0.653	0.003	0.982	0.640	<0.001	50
	Experimental Group 2	0.036	0.804	0.007	0.964	0.719	<0.001	50
Compulsion	Control Group	0.010	0.947	0.038	0.795	0.946	<0.001	50
	Experimental Group 1	0.317	0.025	0.284	0.046	0.833	<0.001	50
	Experimental Group 2	0.040	0.784	-0.220	0.125	0.618	<0.001	50
Social withdrawal	Control Group	-0.086	0.553	-0.098	0.499	0.969	<0.001	50
	Experimental Group 1	0.258	0.070	0.260	0.068	0.976	<0.001	50
	Experimental Group 2	0.134	0.352	0.084	0.560	0.799	<0.001	50
Social aggression	Control Group	0.0541	<0.001	0.0564	<0.001	0.952	<0.001	50
	Experimental Group 1	-0.03	0.838	-0.017	0.908	0.904	<0.001	50
	Experimental Group 2	-0.194	0.177	-0.181	0.209	0.736	<0.001	50
Sexual psychology	Control Group	-0.364	0.009	-0.361	0.010	0.995	<0.001	50
	Experimental Group 1	-0.054	0.708	-0.008	0.956	0.859	<0.001	50
	Experimental Group 2	0.227	0.052	0.070	0.627	0.653	<0.001	50
Dependence	Control Group	0.059	0.684	0.015	0.918	0.814	<0.001	50
	Experimental Group 1	-0.046	0.753	-0.150	0.300	0.698	<0.001	50
	Experimental Group 2	0.076	0.598	0.014	0.923	0.711	<0.001	50
Impulse	Control Group	0.099	0.492	0.026	0.855	0.819	<0.001	50
	Experimental Group 1	-0.174	0.226	-0.352	0.012	0.694	<0.001	50
	Experimental Group 2	0.012	0.933	-0.010	0.943	0.827	<0.001	50
Psychosis	Control Group	0.205	0.153	0.177	0.218	0.977	<0.001	50
	Experimental Group 1	0.147	0.308	-0.906	0.509	0.822	<0.001	50
	Experimental Group 2	-0.243	0.089	-0.189	0.189	0.868	<0.001	50

At level 0.01 (two-tailed), the correlation was significant.

Control Group Analysis.In the Control Group, which did not receive any intervention, the correlation analysis revealed generally low correlations across most mental health indicators. For example, indicators like Somatization (r =0.051 to 0.769) and Depression (r =-0.010 to 0.722) showed weak to moderate correlations between different time points, indicating that mental health status varied over time without any consistent pattern.However, Social Aggression (r=0.0541 to 0.952) and Sexual Psychology (r = -0.364) exhibited relatively higher correlations, suggesting that these particular mental health aspects remained more consistent among participants, despite the lack of intervention.Overall, the findings for the Control Group suggest that, without any structured exercise program, the mental health of participants remained largely unstable, with significant fluctuations over time.

Experimental Group 1 Analysis.For Experimental Group 1, which engaged in self-selected physical activities, the correlations for mental health indicators also varied, but certain indicators showed improved stability compared to the Control Group. Notably, Compulsion ($r = 0.317$ to 0.833) and Social Withdrawal ($r = 0.258$ to 0.976) demonstrated higher correlations, suggesting that self-selected physical activities had a positive effect on maintaining these aspects of mental health.However, other indicators, such as Impulse ($r=-0.174$ to 0.694) and Psychosis ($r=0.147$ to 0.822), displayed low correlations, indicating that these mental health measures were still subject to significant variability and were not substantially improved by the self-selected physical activities.Overall, the results for Experimental Group 1 indicate that while self-selected physical activities provided some benefit in maintaining certain mental health indicators, the effects were not as strong or consistent across all measures.

Experimental Group 2 Analysis.In Experimental Group 2, which participated in the structured low-to-moderate intensity aerobicss exercise, the correlation analysis showed generally higher correlations across most mental health indicators. Social Aggression ($r=0.736$ to 0.995) and Sexual Psychology ($r=0.653$ to 0.995) demonstrated particularly strong correlations, indicating that the aerobicss exercise program had a stabilizing effect on these aspects of mental health.Compulsion ($r = 0.618$ to 0.833) also showed relatively high correlations, suggesting that the structured aerobicss exercise effectively reduced compulsive symptoms and helped maintain mental health stability. However, indicators such as Somatization ($r = -0.251$ to 0.457) and Depression ($r= -0.040$ to 0.451) exhibited weaker correlations, indicating that while the aerobicss exercise had a positive impact, its effect on these particular mental health measures was less pronounced.

The findings suggest that the 12-week aerobicss exercise training program is significantly correlated with the mental health of participants, particularly in Experimental Group 2. The high correlations observed across various mental health indicators in this group indicate that structured aerobicss exercise effectively contributes to the stabilization and improvement of mental health.

Multiple Regression

The regression analysis results for the mental health indicators are summarized in Table 6 The table presented provides a detailed summary of the results from the multiple regression analysis conducted on various mental health indicators.

TABLE 6 Results of Multiple Regression Analysis for Mental Health Indicators.

Indicators	R	R²	F-Value	P-value	B (BMI)	P-value (BMI)	B (WHR)	P-value (WHR)
Somatization	0.337	0.114	9.431	<0.001	0.343	<0.001	11.835	0.023
Anxiety	0.305	0.093	7.516	<0.001	0.25	0.016	15.96	0.004
Depression	0.284	0.081	6.444	0.002	0.406	<0.001	3.649	0.56
Inferiority	0.273	0.075	5.938	0.003	0.43	0.002	9.428	0.204
Paranoia	0.443	0.196	17.900	<0.001	0.318	0.016	37.585	<0.001
Force	0.338	0.114	9.449	<0.001	0.433	<0.001	8.944	0.129
Social Withdrawal	0.305	0.083	7.558	<0.001	0.218	0.019	14.752	0.003
Social Attack	0.313	0.098	7.998	<0.001	0.433	<0.001	9.182	0.153
Psychosexual	0.238	0.057	4.424	0.014	0.098	0.295	13.719	0.007
Dependence	0.457	0.209	19.406	<0.001	0.516	<0.001	16.751	0.002
Impulse	0.273	0.075	5.921	0.003	0.417	<0.001	1.319	0.841
Psychosis	0.124	0.015	1.140	0.323	0.065	0.351	4.252	0.256

1.Significance Analysis.Somatization: The p-values for BMI_Post (<0.001) and WHR_Post (0.023) indicate a significant positive effect of BMI and WHR on somatization symptoms.Anxiety: Both BMI_Post (p=0.016) and WHR_Post (p = 0.004) significantly influence anxiety, suggesting that higher BMI and WHR are associated with increased anxiety levels.Paranoia: WHR_Post has a highly significant positive effect (p<0.001), and BMI_Post also shows significance (p=0.016), indicating that both factors contribute significantly to feelings of paranoia.

2.Non-Significant Analysis.Psychosis: Neither BMI_Post (p = 0.351) nor WHR_Post (p =0.256) has a significant effect, indicating that BMI and WHR do not strongly influence psychosis symptoms.Impulse: WHR_Post is non-significant (p=0.841), suggesting that WHR does not have a notable impact on impulse control, although BMI_Post shows significance.

3.Regression Coefficient Analysis.The regression coefficients (B values) help us understand the direction and magnitude of the relationship between BMI, WHR, and mental health indicators:BMI_Post: For most indicators, such as Dependence (B = 0.516) and Force (B = 0.433), BMI shows a positive relationship, meaning an increase in BMI is associated with higher levels of these mental health issues.WHR_Post: WHR shows a strong positive impact on Paranoia (B = 37.585) and Dependence (B =16.751), suggesting that higher WHR contributes significantly to these conditions.

4.R² Value Analysis.The R² value indicates the proportion of variance in the mental health indicators that can be explained by BMI and WHR:Highest R²: Dependence (R² =

0.209) and Paranoia (R^2 = 0.196) have the highest R^2 values, indicating that a substantial portion of the variance in these indicators is explained by BMI and WHR.Lowest R^2: Psychosis (R^2 = 0.015) shows the lowest R^2, suggesting that the regression model poorly explains the variance in psychosis symptoms.

The multiple regression analysis reveals that BMI and WHR significantly impact various mental health indicators among the subjects. Particularly, indicators such as Paranoia and Dependence are strongly influenced by these factors, as reflected by their high R^2 values and significant regression coefficients.However, certain indicators like Psychosis and Impulse do not show significant relationships, indicating that other factors might be at play or that the influence of BMI and WHR is minimal. These findings suggest that while BMI and WHR are critical factors in understanding mental health outcomes in this population, they may not uniformly affect all aspects of mental health.

Discussion

Overview of Findings

The present study evaluated the impact of a 12-week aerobics exercise program on mental health indicators among obese college students. Using a randomized controlled trial design, the study examined group differences in mental health outcomes across baseline, mid-test, and post-test stages, along with the relationship between exercise adherence and mental improvements. The results demonstrated that structured aerobics exercise significantly improved mental health indicators compared to self-selected physical activities and no intervention. Specifically, Experimental Group 2 showed consistent improvements across somatization, anxiety, depression, and other domains, with several indicators reaching or surpassing the national average by the end of the intervention.

Additionally, multiple regression analysis highlighted the roles of BMI and WHR in predicting mental health outcomes. The findings underscore the potential of structured aerobics exercise as a non-pharmacological intervention for mental health improvement in obese populations, while also suggesting the nuanced impact of obesity metrics (BMI and WHR) on mental health.

Interpretation of Results

Effects of aerobics Exercise on Mental Health

Experimental Group 2 demonstrated significant improvements across nearly all mental health indicators compared to the Control Group and Experimental Group 1. These results align with existing literature that emphasizes the benefits of structured aerobics exercise on mental well-being. The observed reductions in symptoms such as somatization, depression, and inferiority can be attributed to several mechanisms:

Physiological Benefits: aerobics exercise is known to regulate neurotransmitters like serotonin and dopamine, which play critical roles in mood regulation and stress management(26). Improved cardiovascular health may also contribute to enhanced brain oxygenation, promoting cognitive and emotional resilience(27).Mental Mechanisms: Structured aerobics exercise may enhance self-efficacy and body image satisfaction, which are particularly relevant for obese individuals struggling with self-esteem and social stigmatization(28).Social Engagement: Participating in group-based aerobics exercise likely provided opportunities for social interaction and support, which are essential for mitigating feelings of isolation and withdrawal.

The moderate-intensity aerobics exercise in Experimental Group 2 appears particularly effective due to its structured nature, which ensures consistency in frequency, intensity, and duration. This contrasts with the less structured approach in Experimental Group 1, where the variability in self-selected activities may have reduced the overall intervention effectiveness.

Role of BMI and WHR in Mental Health Outcomes

The regression analysis revealed significant relationships between BMI, WHR, and several mental health indicators. Higher BMI and WHR were strongly associated with symptoms such as paranoia and dependence, while their influence on psychosis and impulse control was minimal. These findings suggest that:

BMI and WHR as Stressors: Obesity-related metrics like BMI and WHR may serve as mental stressors, contributing to negative self-perception and social anxiety(29). The cultural emphasis on thinness and ideal body shape in young adults may exacerbate this association.Domain-Specific Impacts: The differential effects of BMI and WHR on various mental health indicators imply that obesity's mental impact is multidimensional. For example, paranoia and dependence, which showed strong associations with WHR, may reflect heightened sensitivity to social stigma and dependence on external validation.Limitations of BMI and WHR: While these metrics are critical for understanding physical health, they may not fully capture the mental complexities of obesity. Other factors, such as lifestyle, personality traits, and coping mechanisms, likely mediate mental health outcomes.

Comparison with Previous Research

The findings are consistent with prior studies demonstrating the positive effects of aerobics exercise on mental health, particularly among individuals with obesity(30). However, this study expands on previous research by incorporating a comprehensive mental health scale (CCSMHS) tailored to Chinese college students, offering a culturally relevant perspective(31). Unlike studies focusing solely on anxiety or depression, the inclusion of 12 dimensions (e.g., inferiority, compulsive tendencies, psychosis) provides a more holistic understanding of mental health in obese populations.

In contrast to some earlier research that reported minimal effects of physical activity on mental health, the strong improvements observed in Experimental Group 2 suggest that the structured, moderate-intensity nature of the intervention is critical for achieving significant outcomes. The lack of substantial improvements in Experimental Group 1 highlights the importance of exercise consistency and supervision.

Implications for Practice

Practical Applications

The findings support the integration of structured aerobics exercise programs into university health promotion strategies for obese students. Such programs can:

Serve as a cost-effective, non-invasive intervention for reducing mental distress.Enhance both physical and mental health simultaneously, addressing the dual burdens of obesity and mental health challenges.Encourage social interaction and community building, fostering a supportive environment for students with obesity.University administrators and healthcare providers should consider offering supervised exercise sessions tailored to students' fitness levels, with an emphasis on accessibility and inclusivity(32).

Policy Implications

Policymakers should prioritize the development of public health initiatives that promote physical activity as a preventive and therapeutic tool for mental health(33). These initiatives could include subsidies for fitness programs, partnerships with educational institutions, and campaigns to reduce stigma around obesity and mental health.

Study Limitations and Future Directions

Study Limitations

Sample Characteristics: The study was conducted at a single university, limiting the generalizability of findings to other populations or cultural contexts.Short Intervention Duration: While 12 weeks provided significant insights, longer-term follow-ups are necessary to evaluate the sustainability of the observed benefits.Self-Report Measures: Reliance on self-reported mental health indicators may introduce bias, as participants may underreport or exaggerate symptoms.

Future Research Directions

Extended Follow-Up: Future studies should include follow-up assessments at 6 months and 1 year post-intervention to determine the long-term effects of aerobics exercise on mental health.Broader Populations: Expanding the study to include diverse age groups, cultural backgrounds, and levels of obesity would enhance the generalizability of findings.Comparative Interventions: Investigating the effects of other exercise modalities, such as resistance training or high-intensity interval training (HIIT), would provide a more comprehensive understanding of the relationship between exercise and mental health(34).

This study highlights the significant mental health benefits of a structured 12-week aerobics exercise program for obese college students. By demonstrating improvements in key mental indicators and exploring the nuanced roles of BMI and WHR, the findings offer valuable insights for designing effective mental health interventions. Structured physical activity, when implemented consistently, represents a promising strategy for addressing the intertwined challenges of obesity and mental health in young adults.

Conclusion

The present study investigated the effects of a 12-week structured aerobics exercise program on the mental health of obese college students, utilizing a randomized controlled trial design to ensure the reliability and validity of the findings. The results provide compelling evidence that moderate-intensity aerobics exercise significantly enhances mental well-being among this population, surpassing the benefits observed from self-selected physical activities and no intervention.

Summary of Key Findings

Significant Improvements in Mental Health Indicators: Participants in Experimental Group 2, who engaged in structured aerobics exercise, exhibited substantial reductions in symptoms across multiple mental health domains, including somatization, anxiety, depression, inferiority, paranoia, compulsion, social withdrawal, aggression, psychosexual issues, dependence, impulsivity, and psychosis. By the end of the intervention, many of these indicators not only improved significantly compared to the Control Group but also aligned with or surpassed national average levels.

Superiority of Structured Exercise Over Self-Selected Activities: While Experimental Group 1, participating in self-selected physical activities, showed some improvements in mental health indicators, the changes were neither as consistent nor as pronounced as those observed in Experimental Group 2. This underscores the effectiveness of structured, supervised exercise programs in eliciting meaningful mental benefits.

Correlations Between Exercise Adherence and Mental Health Outcomes: Higher adherence to the structured aerobics exercise program was positively correlated with greater improvements in mental health indicators. This relationship emphasizes the importance of consistency and commitment to the exercise regimen in achieving optimal mental benefits.

Influence of BMI and WHR on mental Health: Multiple regression analyses revealed that BMI and WHR significantly predict certain mental health outcomes, particularly paranoia and dependence. Higher BMI and WHR were associated with increased symptoms in these areas, highlighting the intertwined relationship between physical and mental health in obese individuals.

Implications of the Study

Theoretical Implications

The findings contribute to the theoretical understanding of how structured aerobics exercise influences mental health among obese college students. The study supports the biopsychosocial model, suggesting that physical activity can lead to mental improvements through physiological mechanisms (e.g., neurotransmitter regulation), mental processes (e.g., enhanced self-efficacy), and social factors (e.g., group cohesion).

Practical Implications

Intervention Design: The success of the structured aerobics exercise program provides a model for designing effective mental health interventions for obese college students. Implementing such programs within university settings can be a feasible and cost-effective strategy to promote mental well-being.Policy Development: The study's outcomes advocate for policymakers to prioritize physical activity programs as integral components of mental health promotion initiatives in educational institutions. Allocating resources to develop and maintain such programs could yield significant long-term benefits for student populations.Holistic Health Approaches: Recognizing the significant impact of BMI and WHR on mental health underscores the need for holistic approaches that address both physical and mental aspects of obesity. Integrating nutritional guidance, mental support, and physical exercise could enhance the overall effectiveness of health interventions.

Final Remarks

This study reinforces the critical role of structured aerobics exercise in enhancing the mental health of obese college students. By demonstrating significant improvements across a wide range of mental indicators, the research provides robust evidence supporting the integration of such programs into university health services. Addressing obesity requires a multifaceted approach, and physical activity emerges as a key component with the dual benefit of improving physical health and alleviating mental distress.The intersection of physical and mental health highlighted in this study underscores the necessity for collaborative efforts among educators, healthcare providers, and policymakers. By fostering environments that promote physical activity and support mental well-being, institutions can significantly contribute to the overall health and success of their students. Continued research and investment in this area hold the promise of mitigating the pervasive challenges associated with obesity and mental health disorders, ultimately leading to healthier, more resilient communities.

Ethics statement

This study protocol has been approved by the Ethics Committee of Universiti Teknologi Malaysia (UTMREC-2024-55), and all participants signed informed consent prior to the study.

Author contributions

Yang Wang: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zainal Abidin B Zainuddin**: Methodology, Formal analysis, Data curation, Conceptualization. **Xiaoxia Yan:** Formal analysis, Data curation, Conceptualization.

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Data Availability Statement

The datasets generated during and/or analyzed during the current study are not publicly available due to ethical and privacy considerations but are available from the corresponding author on reasonable request and in accordance with institutional guidelines.

Conflict of Interest

The authors declare no conflict of interest.

Generative AI statement

The authors declare that no Gen AI was used in the creation of this manuscript.

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